4. Crypt-Arithmetic Problem (Constraint Satisfaction)

```
from itertools import permutations

def solve_crypt_arithmetic():
    words = ["SEND", "MORE", "MONEY"]
    unique_chars = list(set("".join(words)))
    first_letters = {w[0] for w in words}

    for perm in permutations(range(10), len(unique_chars)):
        char_map = dict(zip(unique_chars, perm))
        if any(char_map[ch] == 0 for ch in first_letters): continue

        send = int("".join(str(char_map[c]) for c in "SEND"))
        more = int("".join(str(char_map[c]) for c in "MORE"))
        money = int("".join(str(char_map[c]) for c in "MONEY"))

        if send + more == money:
            print("Solution Found:")
            for char, digit in sorted(char_map.items()):
                print(f"{char}: {digit}")
            print(f"\nEquation verification: {send} + {more} = {money}")
            return

if __name__ == "__main__":
    solve_crypt_arithmetic()
```

main.py

Visualize

Run

```python
import itertools

def solve_cryptarithmetic():
    words = ["SEND", "MORE", "MONEY"]
    # Extract unique characters
    unique_chars = set("".join(words))

    if len(unique_chars) > 10:
        print("Invalid puzzle: More than 10 unique letters.")
        return

    chars = list(unique_chars)

    # Try all digit permutations (0-9) matching the number of unique letters
    for perm in itertools.permutations(range(10), len(chars)):
        char_map = dict(zip(chars, perm))

        # Leading letters cannot be zero
        if char_map['S'] == 0 or char_map['M'] == 0:
            continue
```

Output

```
Solution Found!
  9567  (SEND)
+ 1085  (MORE)
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 10652  (MONEY)

Letter Mapping: {'D': 7, 'Y': 2, 'N': 6, 'S': 9, 'M': 1, 'R': 8, 'O': 0, 'E': 5}
```